OPERATIONS MANAGEMENT

PROCESS STRATEGY

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Process Strategy across the Organization

- Process decisions, such as the amount of customer involvement allowed, are strategic in nature. They should further a company's long-term competitive goals. In making process decisions, managers focus on controlling competitive priorities such as quality, flexibility, time, and cost.

- No service can be provided and no product can be made without a process, and no process can exist without at least one service or product; and a firm can never gain competitive advantage with faulty processes.

- Here, we focus on **process strategy**, which specifies the **pattern of decisions** made in managing processes so that the processes will achieve their competitive priorities, as well as **process analysis**, which is the documentation and **detailed understanding of how work** is performed and how it can be redesigned.

- All parts of an organization including external suppliers and customers across the supply chain need to be involved to ensure that processes are providing the most value to their internal and external customers.

- Process strategy guides a variety of process decisions, and in turn is guided by operations strategy and the organization's ability to obtain the resources necessary to support them.

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- We pay particular attention to ways in which these decisions fit together, depending on factors such as competitive priorities, customer contact, and volume, which in turn lead to two basic change strategies for analyzing and modifying processes, namely: (1) process reengineering and (2) process improvement.
- Both these approaches need process analysis to identify and implement changes.

Three principles concerning process strategy are particularly important:

1. The key to successful process decisions is to make choices that fit the situation and that make sense together. A more effective process is one that matches key process characteristics and has a close **strategic fit**.
2. **Individual processes** are the building blocks that eventually create the firm's whole supply chain. The cumulative effect on customer satisfaction and competitive advantage is huge.
3. Whether processes in the supply chain are performed internally or externally, management must pay particular attention to the interfaces between processes. Dealing with these interfaces underscores the need for **cross-functional coordination**.

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Process Strategy Decisions

- One recurring question in managing processes is deciding *how* to provide services or make products.

- Many different choices are available in selecting human resources, equipment, outsourced services, materials, work flows, and methods that transform inputs into outputs.

- Another choice is which processes are to be done in-house, and which processes are to be outsourced — that is, done outside the firm and purchased as materials and services. This decision helps to define the supply chain.

- Process decisions directly affect the process itself and indirectly the services and the products that it provides. Whether dealing with processes for offices, service providers, or manufacturers, operations managers must consider four common process decisions.

- These four decisions are best understood at the process or sub-process level, rather than at the firm level. We begin by defining four basic process decisions and then we discuss these decisions for both service and manufacturing processes.

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- **Process structure** determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics. A **layout** is the physical arrangement of operations (or departments) relative to each other.

- **Customer involvement** reflects the ways in which customers become part of the process and the extent of their participation.

- **Resource flexibility** is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions.

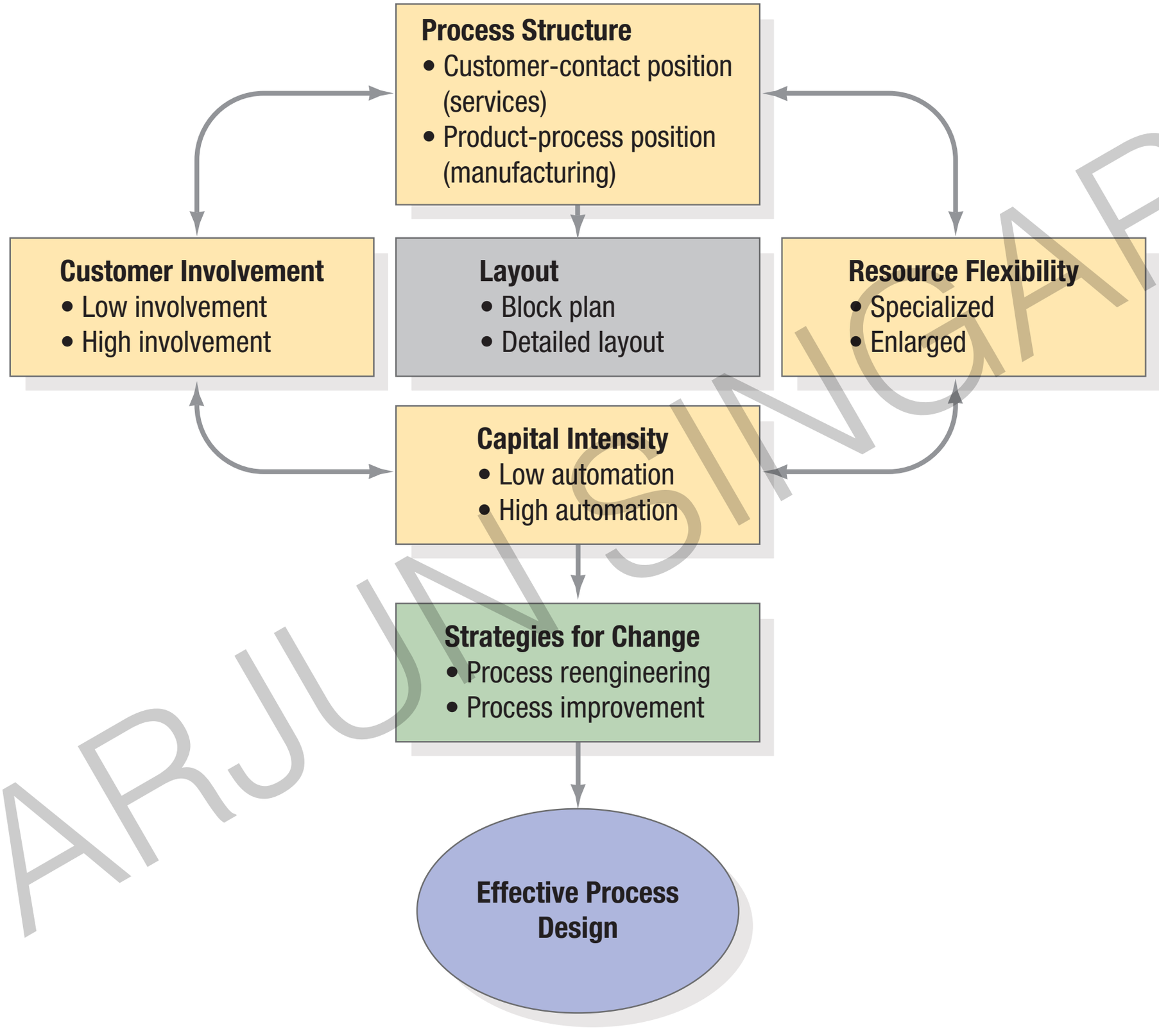
- **Capital intensity** is the mix of equipment and human skills in a process. The greater the cost of equipment relative to the cost of labor, the greater is the capital intensity.

- The concepts that we develop around these four decisions establish a framework within which we can address the appropriate process design in every situation.

- We establish the patterns of choices that create a good fit between the four decisions. For example, if you walk through a manufacturing facility where materials flow smoothly from one work station to the next, you would be tempted to conclude that all processes should be line processes. They seem so efficient and organized.

- However, if volumes are low and the products made are customized, converting to a line process would be a big mistake. When volumes are low and products are customized, resources must be more flexible to handle a variety of products.

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Process Structure in Services

- One of the first decisions a manager makes in designing a well-functioning process is to **choose a process type** that best achieves the competitive priorities for that process.

- Strategies for designing processes can be quite different, depending on whether a service is being provided or a product is being manufactured. We begin with service processes, given their huge implication for workforce resources in industrialized countries.

- A process strategy that gets customers in and out of a fast-food restaurant quickly would not be the right process strategy for a five-star restaurant, where customers seek a leisurely dining experience.

- To gain insights, we must start at the process level and **recognize key contextual variables** associated with the process.

- A good process strategy for a service process depends first and foremost on the type (dimension) and amount of **customer contact**.

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- **Customer contact** is the extent to which the customer is present, is actively involved, and receives personal attention during the service process.

- Face-to-face interaction, sometimes called a *moment of truth* or *service encounter*, brings the customer and service providers together. At that time, customer attitudes about the quality of the service provided are shaped.

- There are **several dimensions of customer contact**. Many levels are possible on each of the five dimensions. Also, some parts of a process can have low contact and other parts of a process can have high contact.

Dimension	High Contact	Low Contact
Physical presence	Present	Absent
What is processed	People	Possessions or information
Contact intensity	Active, visible	Passive, out of sight
Personal attention	Personal	Impersonal
Method of delivery	Face-to-face	Regular mail or e-mail

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Customer-Contact Matrix

- **The customer-contact matrix brings together three elements:** (1) the degree of customer contact, (2) customization, and (3) process characteristics. The matrix is the starting point for evaluating and improving a process.

- **Customer Contact and Customization:** The horizontal dimension of the matrix represents the service provided to the customer in terms of customer contact and competitive priorities. **A key competitive priority is how much customization is needed**.

- The left side of the matrix represents high customer contact, highly customized services, customers more likely to be present & active and a process more likely to be visible to the customer receiving more personal attention.

- The right side of the matrix represents low customer contact, passive involvement, less personalized attention, and a process out of the customer's sight.

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- **Process Divergence and Flow:** The vertical dimension of the customer-contact matrix deals with two characteristics of the process itself: (1) process divergence and (2) flow. Each process can be analyzed on these two dimensions.

- **Process divergence** is the extent to which the process is highly customized with considerable latitude as to how its tasks are performed. If the process changes with each customer, virtually every performance of the service is unique.

- Examples of highly divergent service processes where many steps in them change with each customer are found in consulting, law, and architecture.

- A service with low divergence, on the other hand, is repetitive and standardized. The work is performed exactly the same with all customers, and tends to be less complex. Certain hotel services and telephone services are highly standardized to assure uniformity.

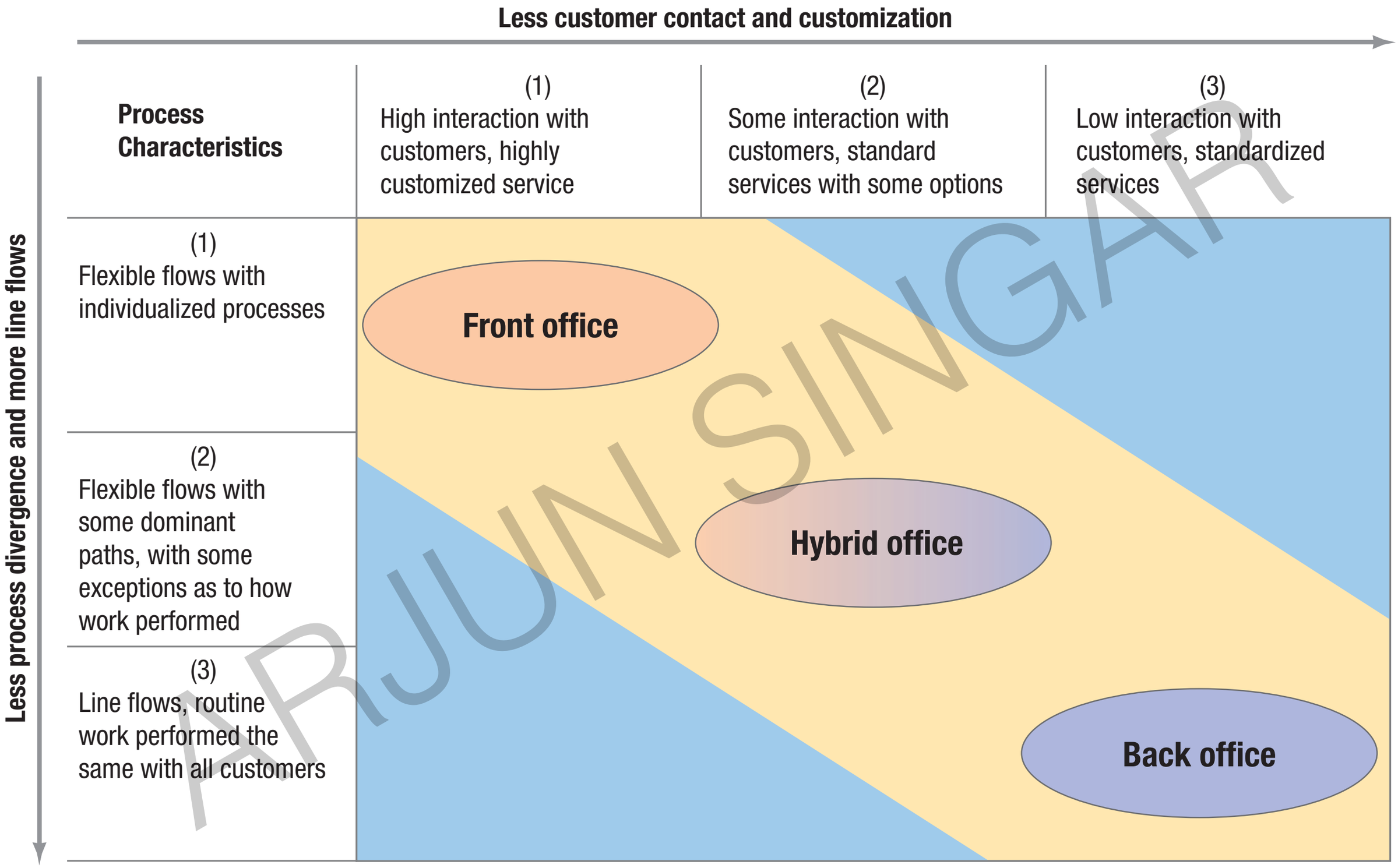
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- Closely related to divergence is how the customer, object, or information being processed flows through the service facility. Work progresses through the sequence of steps in a process, which could range from **highly diverse to linear**.

- When **divergence is high / considerable**, the work flow tends to be more flexible. A **flexible flow** means that the customers, materials, or information move in diverse ways, with the path of one customer or job often criss-crossing the path that the next one takes. Such an appearance goes naturally with high process divergence.

- A **line flow** means that the customers, materials, or information move linearly from one operation to the next, according to a fixed sequence. When **diversity is low and the process standardized**, line flows are a natural consequence.

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Service Process Structuring

- The previous figure also shows several desirable positions in the matrix that effectively connect the service product with the process.

- **The manager here has three process structures, which form a continuum, to choose from:** (1) front office, (2) hybrid office, and (3) back office.

- A **front-office** process has high customer contact where the service provider interacts directly with the internal or external customer. It has high customer contact, high divergence, and a flexible flow. (Ex: Quick service restaurants and ticket booking counters)

- A **hybrid-office** process has moderate levels of customer contact and standard services, with some options available from which the customer chooses. (Ex: Banking, health care and insurance sector)

- A **back-office** process has low customer contact and little service customization. It has low customer contact, low divergence, and a line flow. (Ex: Preparing the monthly client fund balance reports in the financial services industry)

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Process Structure in Manufacturing

- Many processes at a manufacturing firm are actually services to internal or external customers, and so the previous discussion on services applies to them.

- Similarly, manufacturing processes can be found in service firms. Clarity comes when viewing work at the process level, rather than the organizational level.

- Here we focus instead on the manufacturing processes. Because of the differences between service and manufacturing processes, we need a different view on process structure.

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Product-Process Matrix

- **The product-process matrix brings together three elements:** (1) volume, (2) product customization, and (3) process characteristics. It synchronizes the product to be manufactured with the manufacturing process itself.

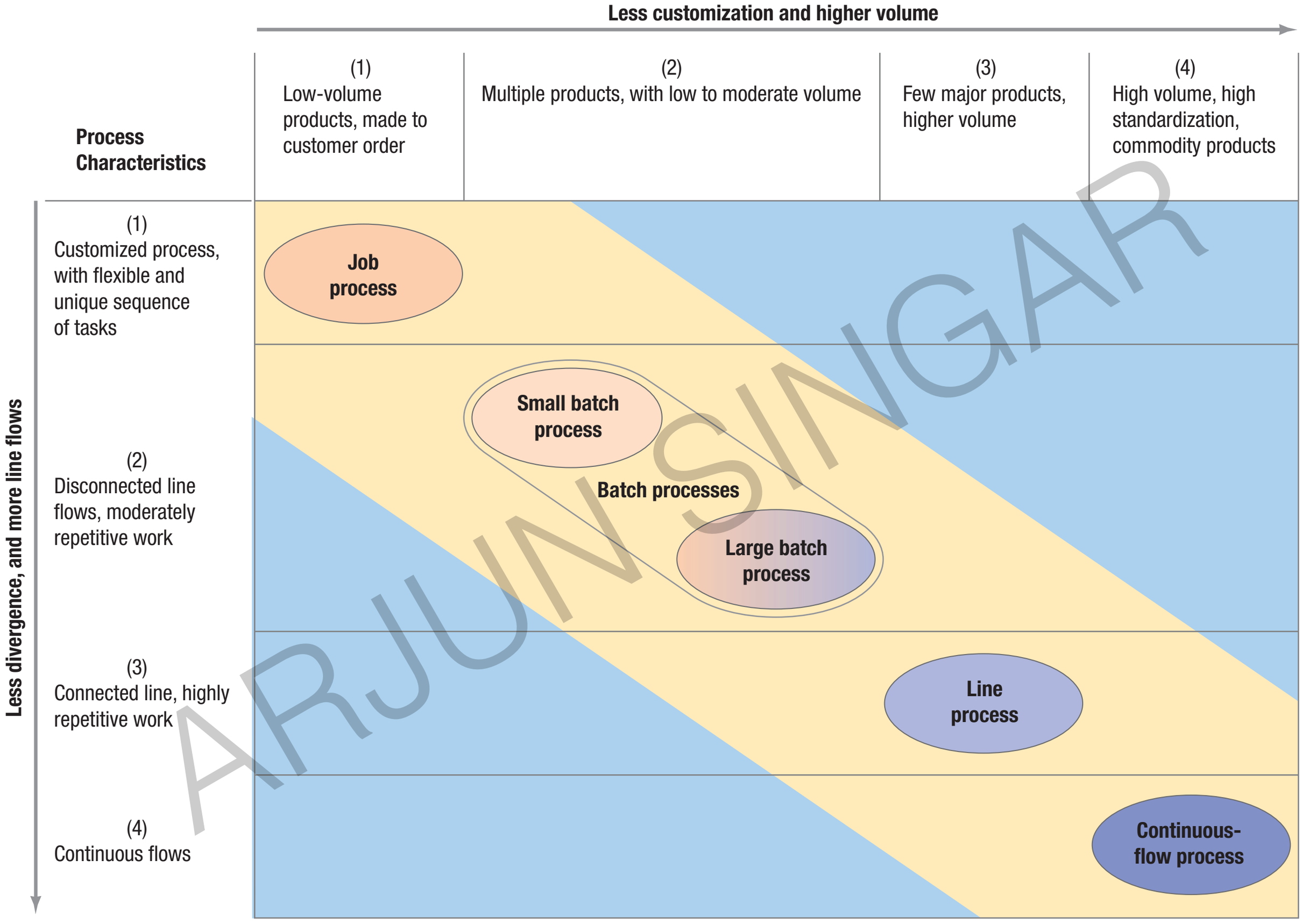
- A good strategy for a manufacturing process depends first and foremost on **volume**.

- Customer contact, a primary feature of the customer-contact matrix for services, normally is not a consideration for manufacturing processes (although it is a factor for the many service processes throughout manufacturing firms).

- For many manufacturing processes, high product customization means lower volumes - **horizontal dimension**, for many of the steps in the process.

- The **vertical dimension** of the product-process matrix deals with the same two characteristics in the customer-contact matrix: **process divergence and flow**. Each manufacturing process should be analyzed on these two dimensions, just as was done for a service process.

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Manufacturing Process Structuring

- The previous figure shows several desirable positions (often called *process choices*) in the product-process matrix that effectively connect the manufactured product with the process.

- **Process choice** is the way of structuring the process by organizing resources around the process or organizing them around the products.

- Organizing around the process means, for example, that all milling machines are grouped together and process all products or parts needing that kind of transformation.

- Organizing around the product means bringing together all the different human resources and equipment needed for a specific product and dedicating them to producing just that product.

- **The manager has four process choices, which form a continuum, to choose from:** (1) job process, (2) batch process, (3) line process, and (4) continuous-flow process.

- The **best choice** for a manufacturing process depends on the **volume and degree of customization** required of the process.

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- A **job process** creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and volume for any one product is low. The workforce and equipment are flexible to handle considerable task divergence. **Examples** are machining a metal casting for a customized order or producing customized cabinets.

- A **batch process** differs from the job process with respect to volume, variety, and quantity. The primary difference is that volumes are higher because the same or similar products or parts going into them are produced repeatedly. A batch process has average or moderate volumes, but process divergence is still too great to warrant dedicating a separate process for each product. The process flow is flexible, but more dominant paths emerge than at a job process, and some segments of the process have a line flow. **Examples** of a batch process are making standard components that feed an assembly line or some processes that manufacture capital equipment, like automobile components.

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- A **line process** lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products. Divergence is minimal in the process or line flows, and little inventory is held between the processing steps. Each step performs the same process over and over, with little variability in the products manufactured. **Examples** of products created by a line process include the assembly of computers, automobiles, appliances, and toys.

- A **continuous flow process** is the extreme end of high-volume standardized production, with rigid line flows. Process divergence is negligible. Its name derives from the way materials move through the process. Usually, one primary material (such as a liquid, a gas, or a powder) moves without stopping through the process. A continuous-flow process differs from a line process in one important respect: Materials flow through the process without stopping until the whole batch is finished. **Examples** of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food.

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Production and Inventory Strategies

- Strategies for manufacturing processes differ from those in services, not only because of low customer contact and involvement, but also because of the ability to use inventories.

- Design-to-order, make-to-order, assemble-to-order, and make-to-stock strategies are four approaches to inventory that should be coordinated with process choice.

- **Design-to-order** strategy is a strategy that involves designing new products that do not currently exist, and then manufacturing them to meet unique customer specifications. Typically a **job process** is employed to create a highly customized product, such as a designer pair of shoes for a particular client.

- **Make-to-order** strategy is a strategy used by manufacturers that make products to customer specifications in low volumes, coupling it with **job or small batch processes**. The processes have high degree of customization and high divergence. Specialized medical equipment, castings, and expensive homes are suited to the make-to-order strategy.

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- **Assemble-to-order** strategy is a strategy for producing a wide variety of products from relatively few subassemblies and components after the customer orders are received. Typical competitive priorities are variety and fast delivery times. The assemble-to-order strategy often involves a **line process for assembly** and a **batch process for fabrication**. Once the specific order from the customer is received, the assembly processes create the product from standardized components and subassemblies produced by the fabrication processes. (**mass customization** - highly divergent process generate wide variety of customized products at reasonably low costs).

- **Make-to-stock** strategy is a strategy that involves holding items in stock for immediate delivery, thereby minimizing customer delivery times. This strategy is feasible for standardized products with high volumes. It is the inventory strategy of choice for **line or continuous-flow processes**. Examples of products produced with a make-to-stock strategy include garden tools, electronic components, soft drinks, and chemicals. Combining a line process with the make-to-stock strategy is sometimes called **mass production**.

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Layout

- A *layout* is the physical arrangement of operations (or departments) created from the various processes and puts them in tangible form.

- An *operation* is a group of human and capital resources performing all or part of one or more processes. For example, an operation could be several customer service representatives in a customer reception area; a group of machines and workers producing cell phones; or a marketing department.

- The flows across departmental lines could be informational, services, or products. Process structures that create more flows across departmental lines, as with job or batch processes, are the most challenging layout problems.

- Here we demonstrate an approach to layout design that positions those departments close together that have strong interactions between them. It involves three basic steps, whether the design is for a new layout or for revising an existing layout: (1) gather information (space requirements, available space, and closeness factors), (2) develop a block plan (allocates space and indicates placement of each operation) and closeness matrix (measure of the relative importance of each pair of operations being located close together), and (3) design a detailed layout.

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Customer Involvement

- Customer involvement reflects the ways in which customers become part of the process and the extent of their participation. It is especially important for many service processes, particularly if customer contact is (or should be) high. Some customers seek active participation in and control over the service process, particularly if they will enjoy savings in both price and time. The manager must assess whether advantages outweigh disadvantages, judging them in terms of the competitive priorities and customer satisfaction. (Example: Self Service Stations & IKEA)

Possible Disadvantages

- Can be disruptive, making the process less efficient
- Managing timing and volume of customer demands can be challenging
- Exposing the facilities and employees can have quality implications
- Requires interpersonal skills which comes at a cost
- Layouts may have to be revised and multiple locations may be necessary

Possible Advantages

- Increased net value to the customer
- Can mean better quality, faster delivery, greater flexibility, and lower cost
- May reduce product, shipping, and inventory costs
- May help coordinate processes across the supply chain
- Processes may be revised to accommodate the customers' role

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Resource Flexibility

- Just as managers must account for customer contact when making customer involvement decisions, so must they account for process divergence and diverse process flows when making resource flexibility decisions. High task divergence and flexible process flows require more flexibility of the process's resources— its employees, facilities, and equipment.

Workforce

- Operations managers must decide whether to have a **flexible workforce**. Members of a flexible workforce are capable of doing many tasks, either at their own workstations or as they move from one workstation to another.

- However, such flexibility often comes at a cost, requiring greater skills and thus more training and education.

- Worker flexibility can be one of the best ways to achieve reliable customer service and alleviate capacity bottlenecks.

- Resource flexibility helps to absorb the feast-or-famine workloads in individual operations that are caused by low-volume production, divergent tasks, flexible flows, and fluid scheduling.

- The type of workforce required also depends on the need for volume flexibility.

- When conditions allow for a smooth, steady rate of output, the likely choice is a permanent workforce that expects regular full-time employment.

- If the process is subject to hourly, daily, or seasonal peaks and valleys in demand, the use of part-time or temporary employees to supplement a smaller core of full-time employees may be the best solution.

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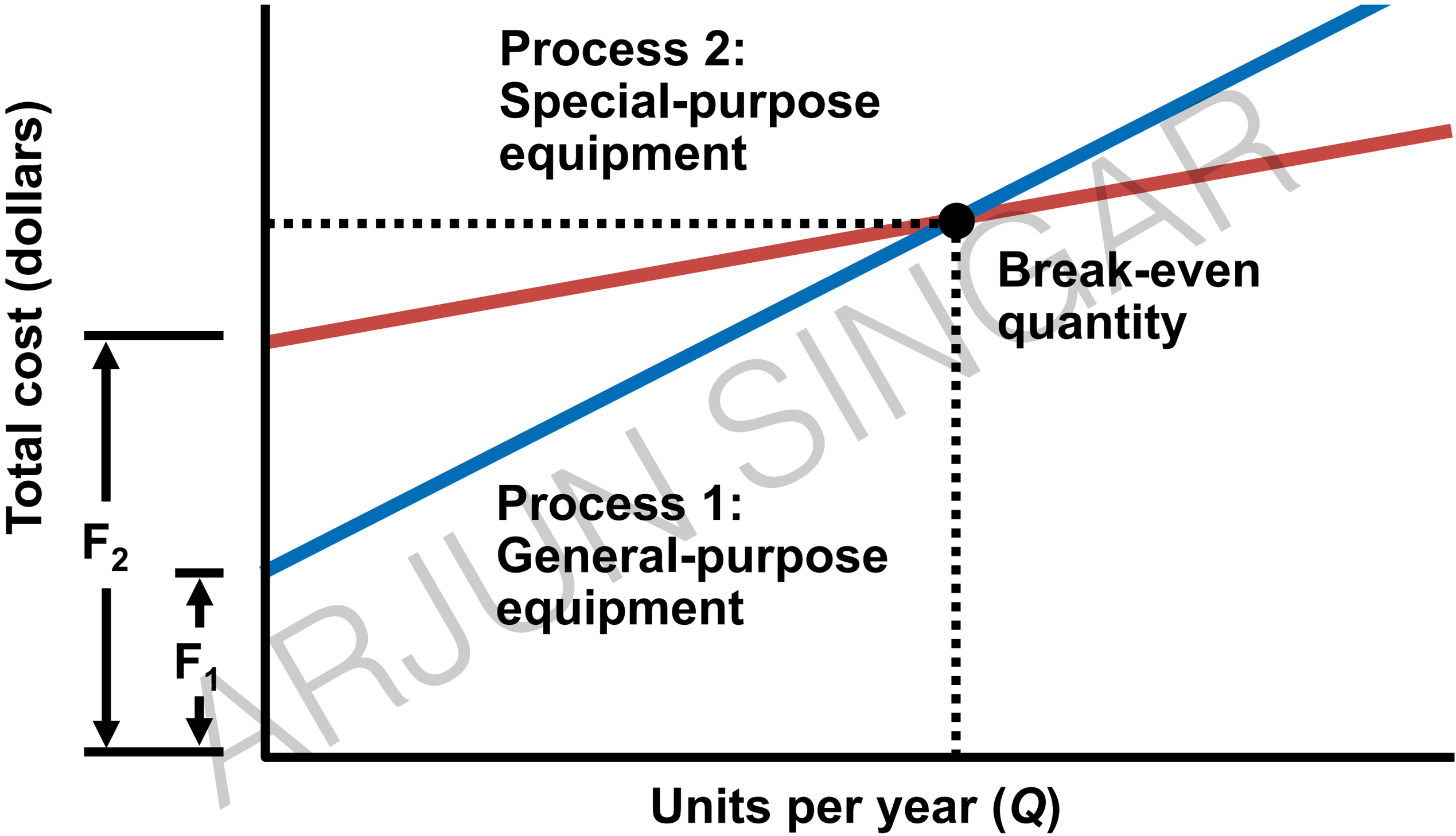
Equipment

- Low volumes mean that process designers should select flexible, general-purpose equipment. Figure illustrates the relationship by showing the total cost lines for two different types of equipment that can be chosen for a process. Each line represents the total annual cost of the process at different volume levels. It is the sum of fixed costs and variable costs. When volumes are low (because customization is high), process 1 is the better choice. It calls for inexpensive general-purpose equipment, which keeps investment in equipment low and makes fixed costs (F_1) small. Its variable unit cost is high, which gives its total cost line a relatively steep slope.

- Conversely, process 2 is the better choice when volumes are high and customization is low. Its advantage is low variable unit cost, as reflected in the flatter total cost line. This efficiency is possible when customization is low because the equipment can be designed for a narrow range of products or tasks. Its disadvantage is high equipment investment and, thus, high fixed costs (F_2). When annual volume produced is high enough, spreading these fixed costs over more units produced, the advantage of low variable costs more than compensates for the high fixed costs.

- The break-even quantity in the figure is the quantity at which the total costs for the two alternatives are equal. At quantities beyond this point, the cost of process 1 exceeds that of process 2. Unless the firm expects to sell more than the break-even amount, which is unlikely with high customization and low volume, the capital investment of process 2 is not warranted.

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Application

BBC is deciding whether to weld bicycle frames manually or to purchase a welding robot. If welded manually, investment costs for equipment are only $10,000. The per-unit cost of manually welding a bicycle frame is $50.00 per frame. On the other hand, a robot capable of performing the same work costs $400,000. Robot operating costs including support labor are $20.00 per frame.

	Welded Manually (Make)	Welded by Robot (Buy)
Fixed costs	$10,000	$400,000
Variable costs	$50	$20

At what volume would BBC be indifferent to these alternative methods?

$$Q = \frac{F_m - F_b}{c_b - c_m} = \frac{\$10{,}000 - \$400{,}000}{\$20 - \$50} = 13{,}000 \text{ frames}$$

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Capital Intensity

- Capital intensity is the mix of equipment and human skills in the process; the greater the relative cost of equipment, the greater is the capital intensity.

- **Automation** is a system, process, or piece of equipment that is self-acting and self-regulating. Although automation is often thought to be necessary to gain competitive advantage, it has both advantages and disadvantages. Thus, the automation decision requires careful examination.

- Substituting labor-saving capital equipment and technology for labor has been a classic way of improving productivity and quality consistency in manufacturing processes.

- If investment costs are large, automation works best when volume is high, because more customization typically means reduced volume.

- One big disadvantage of capital intensity can be the prohibitive investment cost for low-volume operations.

- Generally, capital-intensive operations must have high utilization to be justifiable.

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- Manufacturers use two types of automation: (1) fixed and (2) flexible (or programmable).

- Particularly appropriate for line and continuous-flow process choices, **fixed automation** produces one type of part or product in a fixed sequence of simple operations.

- Operations managers favor fixed automation when demand volumes are high, product designs are stable, and product life cycles are long. These conditions compensate for the process's two primary drawbacks: (1) large initial investment cost and (2) relative inflexibility.

- However, fixed automation maximizes efficiency and yields the lowest variable cost per unit if volumes are high.

- **Flexible (or programmable) automation** can be changed easily to handle various products. The ability to reprogram machines is useful for both low-customization and high-customization processes.

- In the case of high customization, a machine that makes a variety of products in small batches can be programmed to alternate between products.

- When a machine has been dedicated to low customization and a line flow, the machine can simply be reprogrammed with a new sequence of tasks for a new product.

- An **industrial robot**, which is a versatile, computer-controlled machine programmed to perform various tasks, is a classic example of flexible automation.

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Strategic Fit – Decision Patterns

- The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes.

- The choices should fit the situation and each other. When the fit is more *strategic*, the process will be more effective.

- The process structure has a major impact on customer involvement, resource flexibility, and capital intensity.

- Processes can be adjusted for the degree of customization and volume.

- Process flows can be made more or less linear.

- Competitive priorities must be considered when choosing processes.

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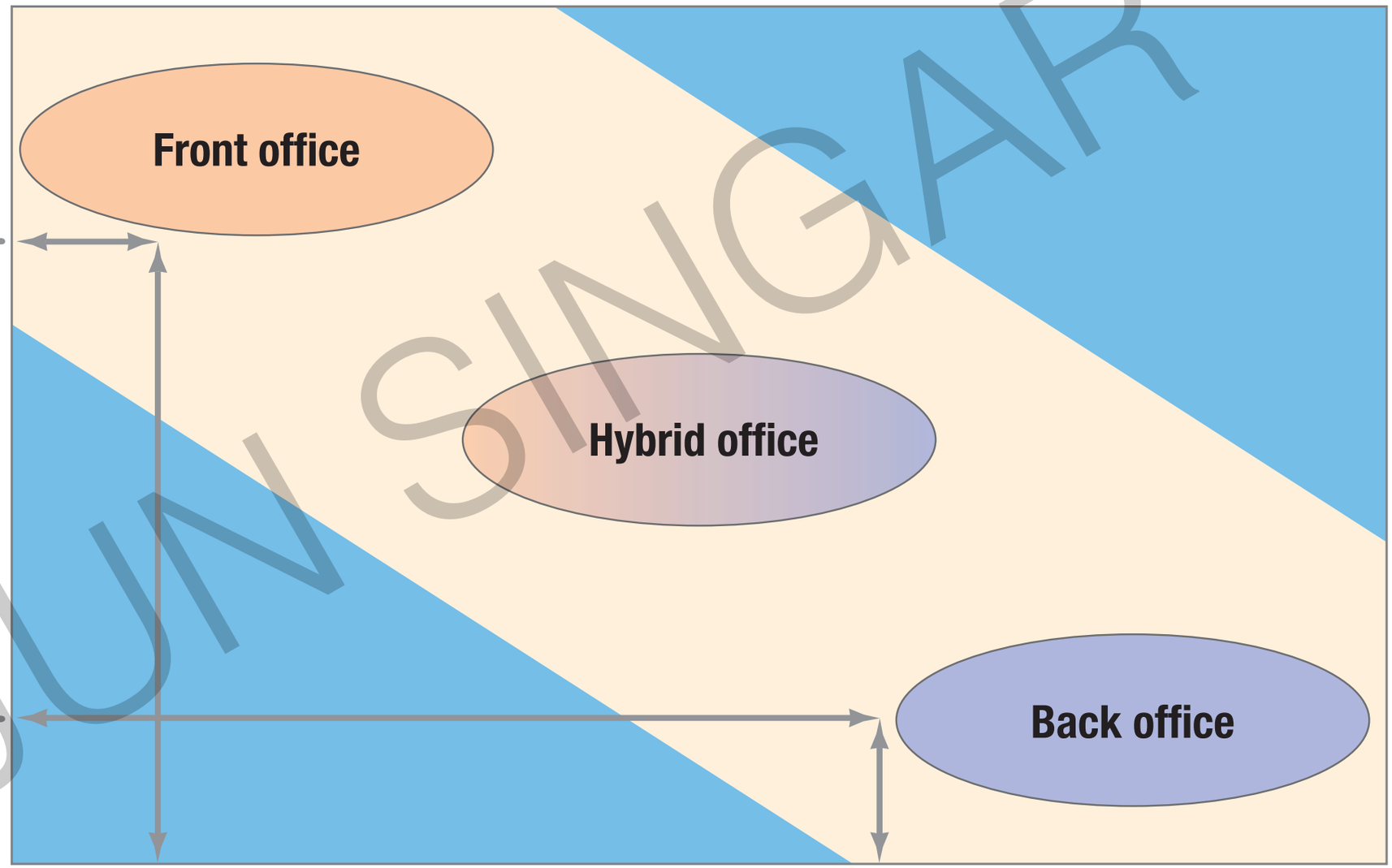
Strategic Fit - Decision Patterns for Service Processes

High customer-contact process

- More divergence, more flexible flows
- More customer involvement
- More resource flexibility
- Capital intensity varies with volume

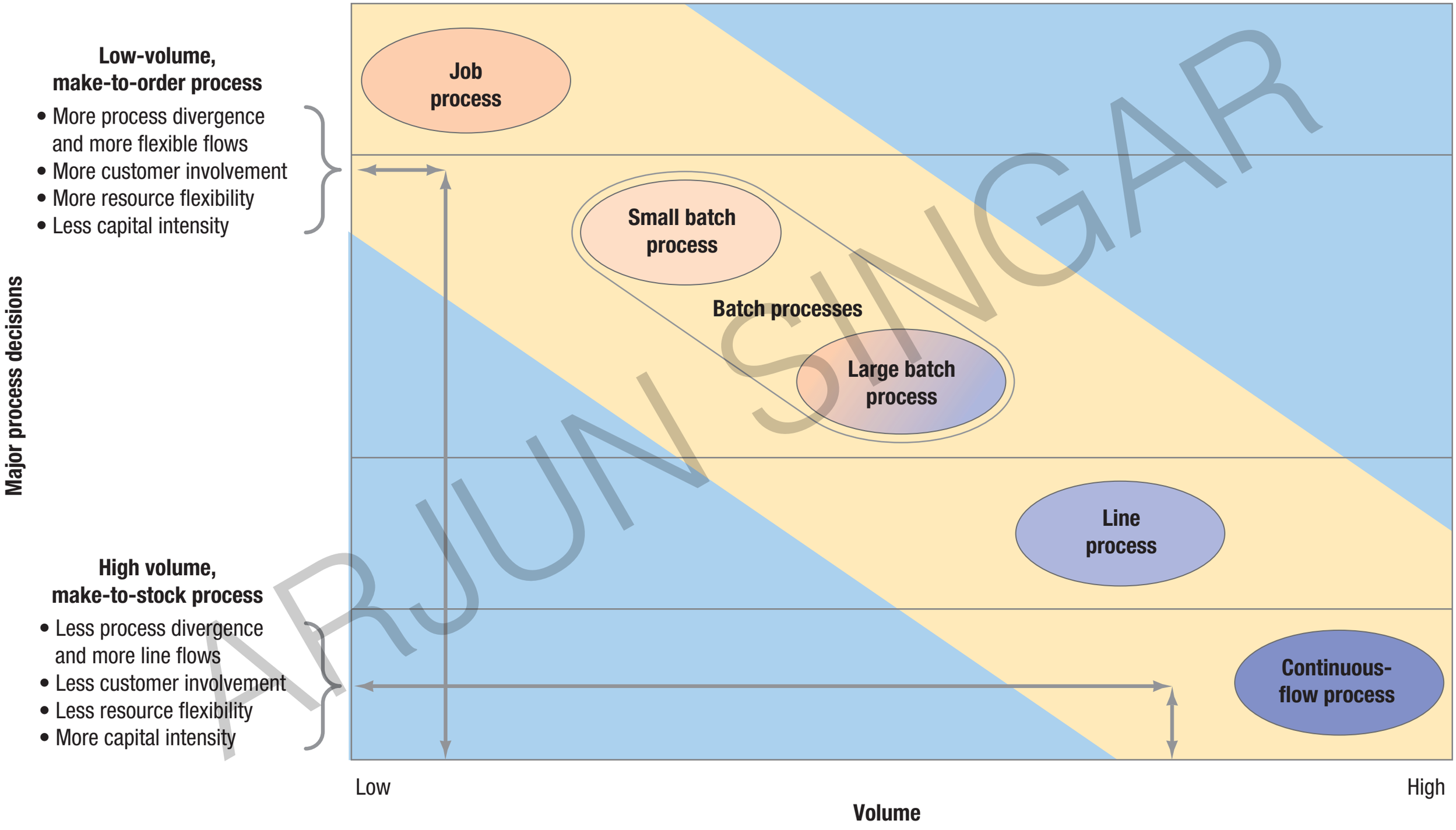
Low customer-contact process

- Less divergence, more line flows
- Less customer involvement
- Less resource flexibility
- Capital intensity varies with volume

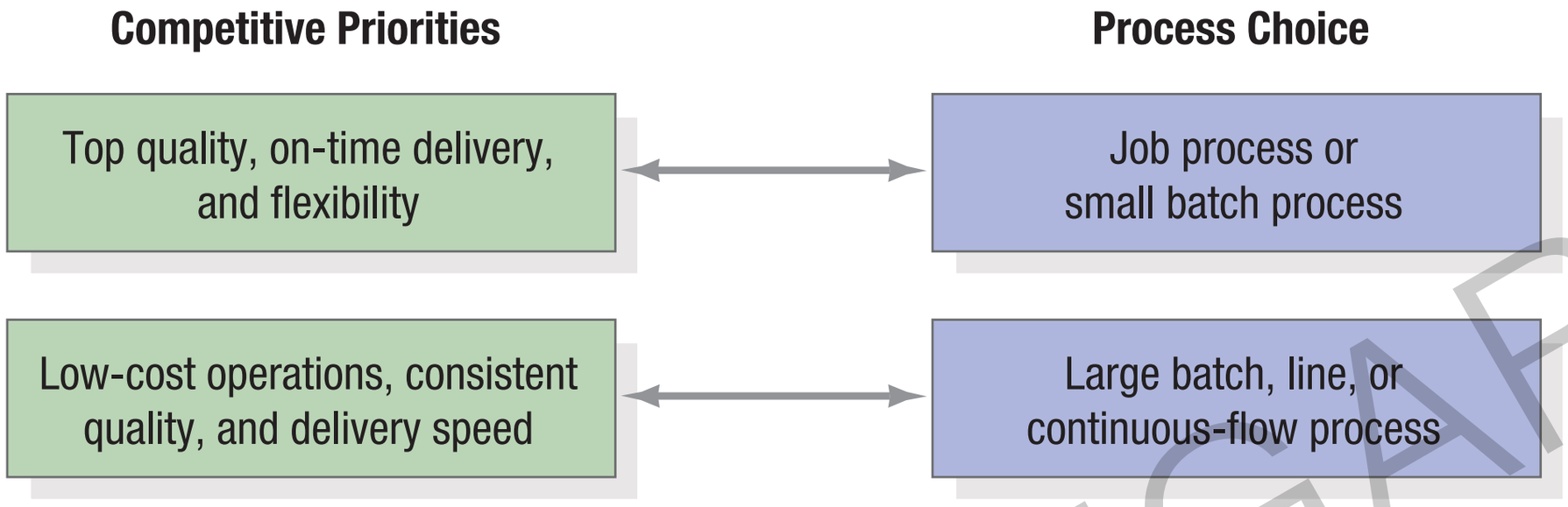


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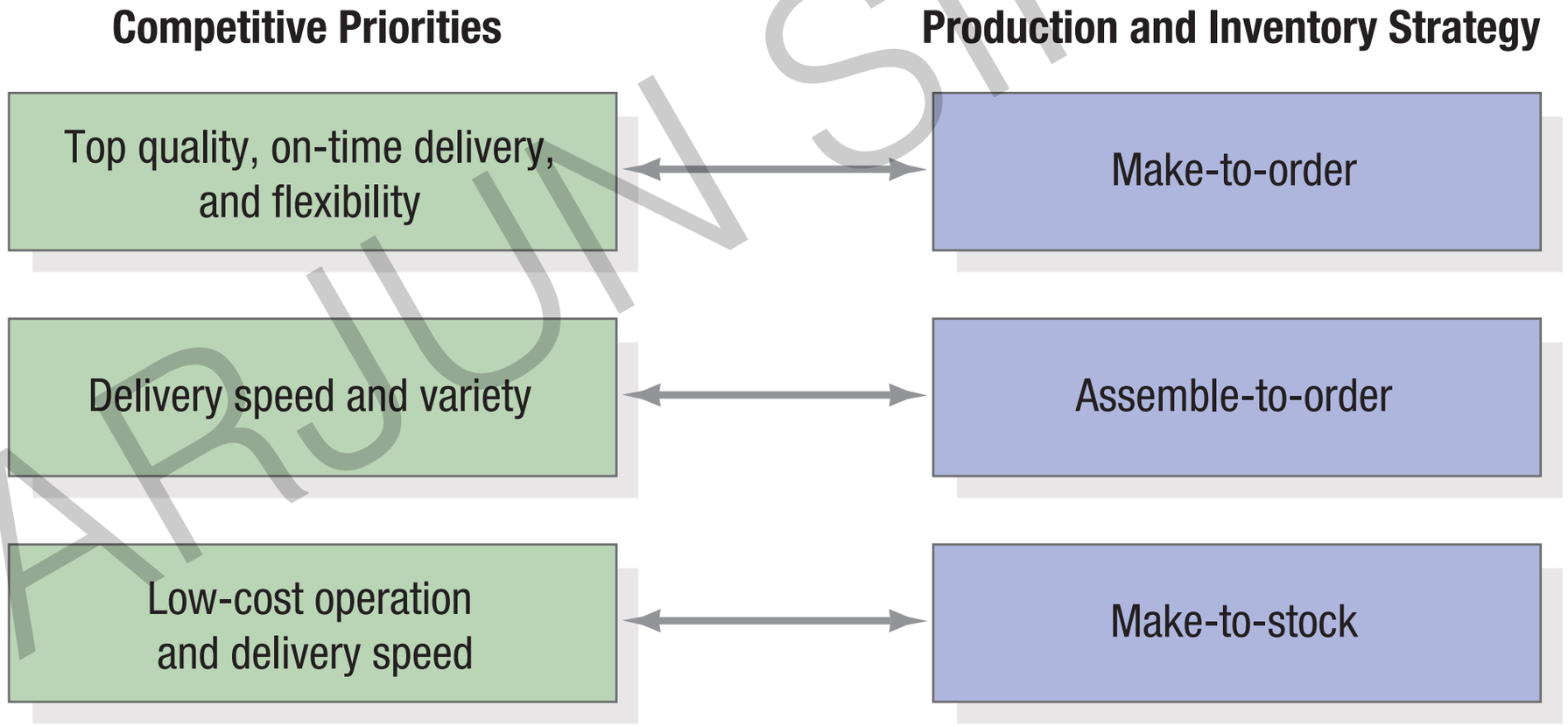
Strategic Fit - Decision Patterns for Manufacturing Processes



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(a) Links with Process Choice



(b) Links with Production and Inventory Strategy

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Strategies for Change

- The four major process decisions represent broad, strategic issues. Decisions that are made must be translated into actual process designs or redesigns. We conclude with two different but complementary philosophies for process design: (1) process reengineering and (2) process improvement.

- **Process reengineering** is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed. Process reengineering is about reinvention, rather than incremental improvement.

- **Process improvement** is the systematic study of the activities and flows of each process to improve it. Its purpose is to "learn the numbers," understand the process, and dig out the details. The relentless pressure to provide better quality at a lower price means that companies must continually review all aspects of their operations. Process improvement goes on, whether or not a process is reengineered. An individual or a whole team examines the process, using the tools. One must look for ways to streamline tasks, eliminate whole processes entirely, cut expensive materials or services, improve the environment, or make jobs safer. One must find the ways to trim costs and delays and to improve customer satisfaction.

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TABLE 3.4 | KEY ELEMENTS OF REENGINEERING

Element	Description
Critical processes	The emphasis of reengineering should be on core business processes. Normal process-improvement activities can be continued with the other processes.
Strong leadership	Senior executives must provide strong leadership for reengineering to be successful. Otherwise, cynicism, resistance ("we tried that before"), and boundaries between departments can block radical changes.
Cross-functional teams	A team, consisting of members from each functional area affected by the process change, is charged with carrying out a reengineering project. Self-managing teams and employee empowerment are the rule rather than the exception.
Information technology	Information technology is a primary enabler of process engineering. Most reengineering projects design processes around information flows, such as customer order fulfillment.
Clean-slate philosophy	Reengineering requires a "clean-slate" philosophy—that is, starting with the way the customer wants to deal with the company. To ensure a customer orientation, teams begin with internal and external customer objectives for the process.
Process analysis	Despite the clean-slate philosophy, a reengineering team must understand things about the current process: what it does, how well it performs, and what factors affect it. The team must look at every procedure involved in the process throughout the organization.

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